

onboard, and its processor employs algorithms to analyse this sound and create the inverted sounds for those noise waves. These inverted sounds are called anti-noise, and they are outputted along with music or other sounds from the input source. The outputted anti-noise cancels ambient noise, and only the music sound reaches your ears.

Types of ANC

In the Feedforward ANC setup, microphones are placed on the outside of the earcups or earbuds. These external microphones capture ambient noise before it reaches your ears. Using digital signal processing, the system generates an opposite sound wave to cancel the noise. Feedforward ANC works best against steady mid- and low-frequency sounds like engine hums or air conditioners.

The Feedback ANC setup, meanwhile, uses microphones placed inside the earcups or earbuds, close to your ears. These internal microphones detect the noise that actually reaches your ear, including any sound that wasn't cancelled properly. So, the underlying processor has to work in real time to analyse and quickly create appropriate anti-noise. This ANC can be sensitive to the earphones' fit and movement within the ear.

And finally, we have modern headphones with a combination of Feedback and Feedforward ANC and are thus called Hybrid ANC. Since it cancels a wider range of noise frequencies, hybrid ANC tends to be more effective. However, the extra processing power and advanced algorithms make it more complex and often more expensive.

Besides these three ANC types, you will also come across terms like Adaptive noise cancellation and Adjustable ANC on the web. Adaptive Noise Cancellation is similar to Hybrid ANC in the sense that it employs algorithms for real-time processing and cancellation of noise. Meanwhile, the Adjustable ANC allows you to increase

or decrease the noise cancellation level through the companion app or physical controls on the headphones.

ENC: How is it different from ANC

While ANC reduces all surrounding noise for the listener, ENC focuses on filtering out most of the background noise so the user's voice sounds crystal clear to the recipient.

Expected noise cancellation

ANC headphones should be able to block low-frequency sounds such as engine noise or air conditioning hum. High-frequency noises, including human speech, may not be fully eliminated.

Fit and comfort

The headphone fit impacts passive isolation. Over-ear designs generally

Feature	Active noise cancellation (ANC)	Environmental noise cancellation (ENC)
Primary purpose	Affects what you hear	Affects what others hear
How it works	Uses external and internal mics to create inverse sound waves that cancel noise.	Uses beamforming mics and algorithms to identify and suppress non-speech noise in the mic signal.
Main benefit	Cleaner listening experience, better focus in noisy environments.	Clearer voice during calls and voice recordings.

So, if you want to block ambient noise for music or concentration, you want ANC, and if you need clearer voice calls, you need ENC.

How to choose ANC headphones

Your budget

Your budget determines the level of ANC performance and features.

offer better isolation and comfort, which is important for long listening sessions.

Battery life

Check the battery life with ANC activated. ANC can drain the headphones' battery, and so, you want something that lasts long so that you get uninterrupted use, especially during travel or extended listening sessions.



High-end models like Sony WH-1000 series, Apple AirPods Max, Bose QuietComfort series, and Sennheiser MOMENTUM 4 Wireless offer advanced algorithms and hardware for superior noise cancellation.

Can I adjust the level of the ANC?

Yes, select headphones like the Sony WH-1000XM3 and XM4 have Noise Cancellation Optimiser to adjust the level of ANC you get. **d**

FAQs

Can ANC and ENC be used together?

Yes, modern earphones and headphones implement both ANC and ENC, but they work independently.